

STATEMENT OF RESEARCH INTERESTS

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Research Focus

My research focuses on trustworthy multimodal AI, especially how AI systems understand and moderate harmful content across text, images, and speech. Online content is increasingly multimodal, yet many safety systems still analyze one modality or one message at a time. This is a serious limitation when harmful meaning comes from the interaction between an image and its caption, speech in a video, text embedded in an image, or a longer conversation.

My work addresses these problems through multimodal content analysis, hate speech and propaganda detection, vision-language models, and large-scale social media analysis. Much of my recent work has used Arabic social media as a challenging testbed, but the underlying questions are broader: how can models recognize harm across modalities, remain reliable across different communities and contexts, and provide decisions that people can understand and verify? My future research will build on this foundation while moving toward general multimodal safety, reasoning, and agentic systems.

Research Foundation

I have worked across the full research pipeline: collecting social media data, building annotated datasets, developing models, organizing shared tasks, and deploying analytical systems. This combination has shaped how I think about AI research. Strong benchmark performance is useful, but a model becomes much more valuable when it can work with noisy real-world data, explain its decisions, and fit into a reliable human workflow.

A major part of this work is the MARSAD Social Media Observatory, a platform for large-scale collection and analysis of online discourse. MARSAD integrates capabilities such as sentiment analysis, propaganda and harmful-content detection, fact-checking signals, topic analysis, and network exploration. Building this system has given me practical experience with scalable model inference, multimodal data, OCR, and analyst-facing interfaces.

I have also developed datasets and benchmarks for hate speech, harmful prompts, stance, propaganda, and multimodal memes. The So Hateful! dataset (LREC-COLING 2024), for example, provides fine-grained annotations for Arabic hate speech and has received 40 Google Scholar citations since publication. My work on ArAIEval, MAHED, ImageEval, AHA-Memes, and ArGuard has further explored multimodal and safety-related problems. I have organized or co-organized several international shared tasks, which has allowed me to build evaluation resources and work with a broad research community. My Google Scholar profile currently reports 862 citations, an h-index of 14, and an i10-index of 19.

Future Research: Trustworthy Multimodal AI

My future work will focus on three connected directions. The first two extend directly from my current research. The third is a longer-term direction that I want to develop as multimodal systems become more capable of using tools and acting on complex information.

1. Multimodal Content Safety and Moderation

Harmful online content is rarely limited to clean text. A meme may become hateful only when its image and caption are interpreted together. A video may contain harmful speech, visual symbols, overlaid text, and comments that change its meaning. I want to develop moderation models that jointly analyze text, images, video, and speech rather than treating these signals separately.

This direction will combine vision-language models with technologies such as automatic speech recognition (ASR) and OCR to understand real-world multimedia. I am particularly interested in implicit hate, coded language, propaganda, and other cases where surface-level classification is not enough. The goal is to build

systems that remain useful under noisy inputs, changing contexts, and different forms of multimodal manipulation.

2. Vision-Language Models and Multimodal Reasoning

A second direction is to move from detection toward reasoning. Current vision-language models can produce strong predictions while still relying on weak or misleading evidence. For safety-sensitive applications, it is not enough to know that a model assigned a harmful label. We also need to know what evidence it used and whether that evidence supports the decision.

I plan to study models that reason across visual, textual, and speech information and connect their decisions to specific evidence. For example, a model should be able to identify which visual element, spoken phrase, caption, or contextual relationship makes a piece of content harmful. I am interested in evaluating both the final prediction and the quality of the reasoning behind it. This can make multimodal moderation more transparent and easier for human reviewers to audit.

3. Toward Agentic Multimodal Systems

A longer-term direction of my research is agentic AI. I do not see agents as a separate topic from my current work, but as a way to extend multimodal analysis beyond a single model. Complex content-safety decisions may require several capabilities: extracting speech with ASR, reading text with OCR, interpreting images with a vision-language model, retrieving background information, checking claims, and applying specialized safety models.

I want to investigate systems that can decide which tools are needed, gather evidence from different sources, reason across their outputs, and produce a traceable conclusion. A key research question is when this agentic approach improves reliability over a fixed pipeline or a single large model. I am equally interested in its failure modes: unnecessary tool use, weak evidence, cascading errors, and overconfident decisions. This makes content safety a useful setting for studying agentic systems because both the benefits and risks can be measured clearly.

Research Goal

These directions share one goal: to move from models that simply classify content toward systems that can perceive, reason, and act on complex multimodal information responsibly. My previous work in content analysis and multimodal safety provides the foundation; vision-language reasoning is the next methodological step; and agentic systems offer a longer-term path toward more flexible AI. I want to develop these ideas in a research environment where multimodal learning, trustworthy AI, and real-world deployment can inform each other.